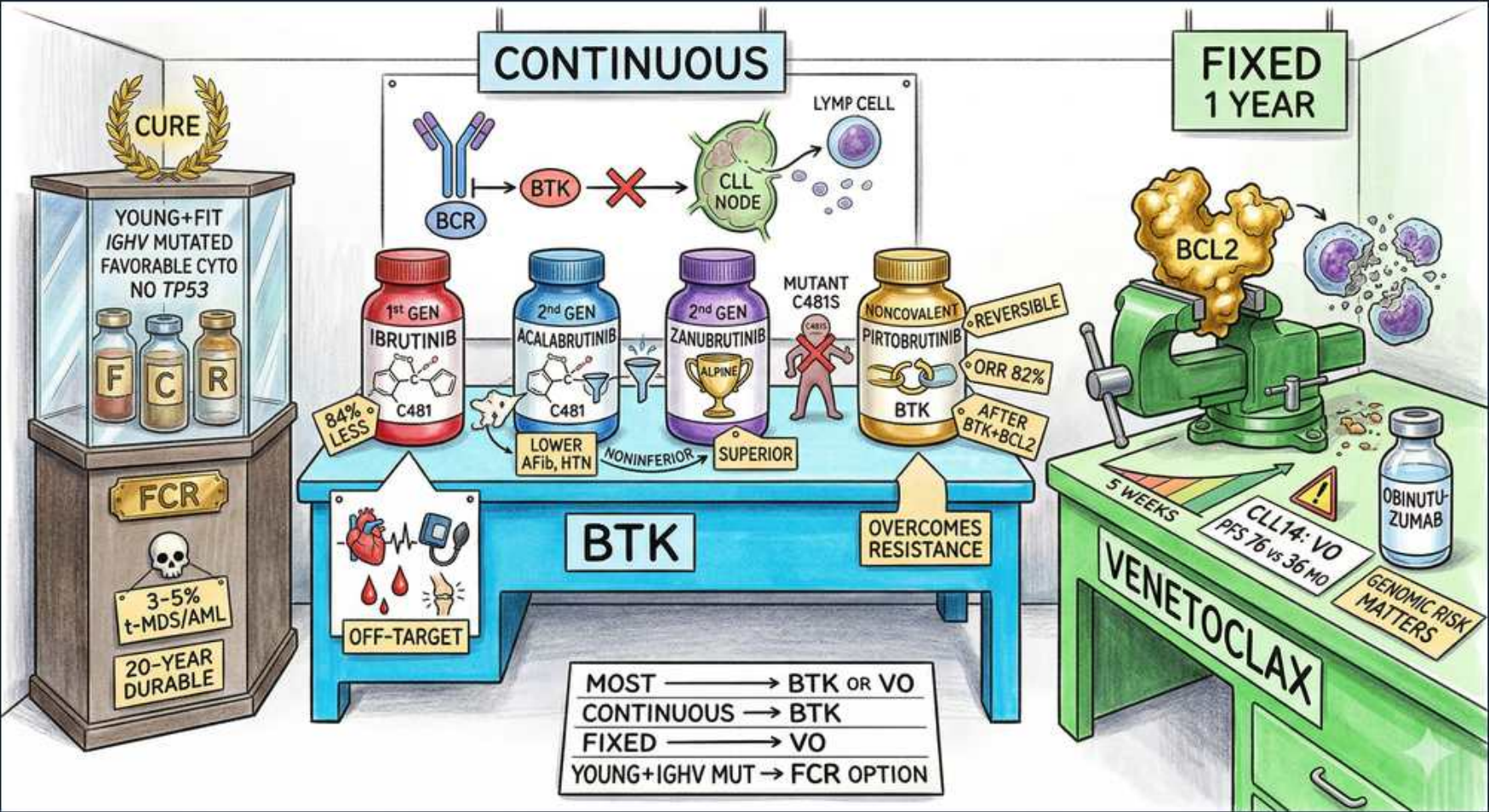


CLL — BTK Inhibitors, Venetoclax & Treatment Selection



1. BCR → BTK Pathway

WHAT YOU SEE

BCR receptor, BTK arrow blocked, X over CLL survival signal

WHAT IT ENCODES

B-cell receptor (BCR) signaling drives CLL survival through Bruton tyrosine kinase (BTK). BTK inhibitors block this pathway, causing nodal shrinkage and a transient redistribution lymphocytosis as cells leave nodes.

BOARD TRAP

The early lymphocytosis spike after starting a BTK inhibitor is REDISTRIBUTION, not progression — don't stop the drug thinking it's failing.

2. Ibrutinib (1st Gen BTK)

WHAT YOU SEE

Red '1st GEN IBRUTINIB' bottle, C481 binding, off-target effects

WHAT IT ENCODES

Ibrutinib is the first-generation covalent BTK inhibitor binding cysteine-481. Highly effective but off-target kinase inhibition (EGFR, ITK, TEC) causes atrial fibrillation, hypertension, bleeding, and diarrhea.

BOARD TRAP

Ibrutinib's cardiac toxicity (afib, HTN, sudden death) is its Achilles heel — bleeding risk means holding it around surgery and caution with anticoagulants.

3. Acalabrutinib (2nd Gen)

WHAT YOU SEE

'2nd GEN ACALABRUTINIB' bottle, C481, lower afib/HTN

WHAT IT ENCODES

Acalabrutinib is a more selective 2nd-gen covalent BTK inhibitor (still C481-binding) with fewer off-target effects — less atrial fibrillation and hypertension than ibrutinib (ELEVATE-RR non-inferior, safer).

BOARD TRAP

Acalabrutinib absorption is impaired by proton pump inhibitors (pH-dependent) — separate from PPIs/antacids. Headache early on is common but self-limited.

4. Zanubrutinib (2nd Gen)

WHAT YOU SEE

'2nd GEN ZANUBRUTINIB' bottle, SUPERIOR label

WHAT IT ENCODES

Zanubrutinib is a 2nd-gen covalent BTK inhibitor that showed SUPERIORITY (not just non-inferiority) to ibrutinib in PFS (ALPINE) with less cardiac toxicity — now a preferred BTK option.

BOARD TRAP

Zanubrutinib beat ibrutinib on efficacy AND safety — don't assume all 2nd-gen agents are merely 'safer but equal'; zanubrutinib improved PFS.

5. Pirtobrutinib (Non-covalent)

WHAT YOU SEE

'NONCOVALENT PIROBRUTINIB' bottle, reversible, ORR 82%, after BTK/BCL2

WHAT IT ENCODES

Pirtobrutinib is a non-covalent, reversible BTK inhibitor that binds independent of C481 — active in CLL that progressed on covalent BTK inhibitors due to C481S mutation (BRUIN ORR ~82%).

BOARD TRAP

Non-covalent pirtobrutinib overcomes C481S resistance — the very mutation that defeats ibrutinib/acalabrutinib/zanubrutinib. Use it after covalent BTK failure.

6. C481S Resistance Mutation

WHAT YOU SEE

'MUTANT C481S' red figure, BTK resistance

WHAT IT ENCODES

C481S is the dominant resistance mutation to covalent BTK inhibitors — it mutates the cysteine the drugs bind, abolishing covalent attachment. Detectable before clinical relapse.

BOARD TRAP

C481S confers cross-resistance to ALL covalent BTK inhibitors (ibrutinib/acala/zanu) — switching among them fails; need non-covalent pirtobrutinib or venetoclax.

7. Venetoclax (BCL2 Inhibitor)

WHAT YOU SEE

Green 'VENETOCLAX' machine/vial, BCL2 target, CLL14 trial, fixed 1 year

WHAT IT ENCODES

Venetoclax is an oral BCL2 inhibitor (BH3-mimetic) that restores apoptosis. Venetoclax-obinutuzumab (CLL14) gives fixed-duration 12-month therapy with deep MRD-negative remissions, including TP53-aberrant disease.

BOARD TRAP

Venetoclax is FIXED-duration (time-limited), unlike continuous BTK inhibitors — a key distinction. Requires TLS prophylaxis with dose ramp-up.

8. Tumor Lysis Syndrome Risk

WHAT YOU SEE

Venetoclax ramp-up, 'WEEKS' gradual schedule

WHAT IT ENCODES

Venetoclax causes tumor lysis syndrome — mandatory 5-week weekly dose ramp-up (20→50→100→200→400 mg), hydration, allopurinol/rasburicase, and labs. High-burden patients hospitalized for first doses.

BOARD TRAP

Never start venetoclax at full dose — rapid lysis of bulky CLL can cause fatal hyperkalemia/AKI. The slow ramp is non-negotiable for TLS prevention.

9. FCR — Curative Niche

WHAT YOU SEE

'CURE' trophy cabinet: YOUNG+FIT, IGHV MUTATED, FAVORABLE CYTO, NO TP53

WHAT IT ENCODES

FCR remains an option offering functional cure ONLY in young, fit patients with IGHV-mutated, favorable cytogenetics, no TP53 abnormality. Achieves ~20-year durable remissions in this subset.

BOARD TRAP

FCR carries a 3-5% risk of therapy-related MDS/AML — a long-term toxicity targeted agents avoid. Reserve strictly for the favorable-biology young patient.

10. FCR Secondary MDS/AML Risk

WHAT YOU SEE

Skull cabinet: '3-5% t-MDS/AML', 20-year durable

WHAT IT ENCODES

FCR's fludarabine/cyclophosphamide cause a 3-5% risk of therapy-related myeloid neoplasms — the trade-off against its long remissions. This risk is a major reason targeted agents are now preferred.

BOARD TRAP

The leukemogenic risk of FCR is why it's avoided in older/unfit patients and in anyone who can use safer targeted therapy with similar efficacy.

11. Treatment Algorithm Summary

WHAT YOU SEE

Bottom legend: MOST→BTK or VO; CONTINUOUS→BTK; FIXED→VO; YOUNG+IGHV MUT→FCR

WHAT IT ENCODES

Most CLL: choose continuous BTK inhibitor OR fixed-duration venetoclax-obinutuzumab (VO). Continuous = BTK; fixed = VO. FCR reserved for young IGHV-mutated. TP53-aberrant → targeted only.

BOARD TRAP

Chemoimmunotherapy is no longer first-line for most CLL — the board answer is a targeted agent. Reserve FCR for the narrow favorable-biology niche.

12. Continuous vs Fixed Duration

WHAT YOU SEE

'CONTINUOUS' header over BTK table vs 'FIXED 1 YEAR' over venetoclax

WHAT IT ENCODES

BTK inhibitors are given CONTINUOUSLY until progression/intolerance; venetoclax-obinutuzumab is FIXED 12-month duration. Choice balances patient preference, comorbidities (cardiac → avoid ibrutinib), and adherence.

BOARD TRAP

Don't stop a BTK inhibitor at an arbitrary timepoint — it's continuous. Conversely, venetoclax-obinutuzumab is meant to be stopped at 1 year, not continued indefinitely.
